

Reproducible Detection of AI-Generated Faces via Open Handcrafted Features on PCA-Residual Images

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Abstract

Recent work has shown that removing the dominant principal components from a face image and examining what remains, the PCA residual, exposes a distributional gap between real photographs and AI-generated faces that a Kolmogorov-Smirnov test can detect using a proprietary fractal-descriptor toolkit. That finding is compelling but has two practical limits: the descriptor extraction depends on closed-source software that other researchers cannot inspect, install, or rerun, and the analysis stops at showing that two distributions differ, without ever training a classifier that outputs a real-or-fake decision for a new image. This paper closes both gaps. Each grayscale face image is reduced to its residual after removing the top thirty-two row-wise principal components computed independently for that image, and three descriptors built entirely from openly documented, installable methods, a log-magnitude frequency histogram, a rotation-invariant local binary pattern histogram, and a Gaussian high-pass noise histogram, are computed on that residual and concatenated into a 187-dimensional feature vector. Five classical classifiers are trained on this representation over 40,000 balanced grayscale faces, 20,000 real images drawn from FFHQ and 20,000 StyleGAN-generated images, split 80/10/10 with a fixed random seed and no leakage between the fitted scaler and the held-out data. On the test set, the best-performing gradient-boosted tree separates real from fake with an accuracy of 86.9 percent and an area under the ROC curve of 0.944, without any deep network and without any component that a third party cannot install from a public package index. The paper reports the complete metric table across all five classifiers, walks through every design decision behind the feature pipeline in enough detail to defend under direct questioning, situates the approach against fifteen related lines of work spanning fractal, frequency, texture, and noise-based detection, and states plainly which extensions, a sensitivity sweep over the residual truncation level, a controlled ablation isolating the PCA-residual step itself, and robustness testing under common perturbations, remain open rather than presenting them as settled. Every number in this paper comes from code that was executed end to end on the stated data, and the full pipeline was independently re-run once to confirm the results reproduce.

Keywords: AI-generated face detection; PCA residual; handcrafted features; local binary patterns; frequency-domain analysis; gradient boosting; reproducibility

1 Introduction

Face images synthesized by generative adversarial networks and diffusion models have become difficult to distinguish from photographs by eye. The standard response in the detection literature has been to train deep convolutional or transformer networks directly on pixels, using benchmarks such as FaceForensics++ [3] to compare methods. These detectors reach high in-distribution accuracy, but they are expensive to train,

opaque in what they key on, and, as a class, they are not the only viable approach: a parallel line of work argues that hand-designed descriptors, computed once and fed to a classical classifier, can reach competitive accuracy while remaining cheap to train, fast at inference, and open to direct inspection of what feature is driving which decision [4, 9, 10, 14].

Within that second line of work, a recent direction analyzes not the face image itself but its residual after a dominant structure is removed by principal component analysis. The originating study in this direction generates the residual by discarding the leading principal components of an image and shows, via a Kolmogorov-Smirnov test over fractal descriptors, fractal dimension, multifractal spectrum width, and lacunarity, computed on that residual, that real and AI-generated faces occupy measurably different distributions [1]. This is a genuine and useful finding: it establishes that the PCA-residual representation carries a real/fake signal. It also has two limitations that motivate the present paper. First, the fractal descriptors in [1] are extracted with FreeAeon, a closed-source toolkit. A reader cannot install it, inspect its internals, or verify independently that the reported fractal values are computed the way the paper describes. Second, and more fundamentally, [1] never trains a classifier. Its evidence is a distributional test showing that two populations of fractal values differ. It does not report an accuracy, an F1 score, or an AUC for the task an actual detector must perform, deciding whether one new image is real or fake.

This raises a direct question: does the PCA-residual representation, shown by [1] to carry a real/fake signal under fractal descriptors, still carry that signal when paired with open, standard descriptors instead of a closed-source fractal toolkit, and can that signal be turned into an actual classifier with reportable accuracy? A second question follows immediately: among the classifiers commonly used with fixed-length handcrafted feature vectors, which family, a linear model, bagged trees, or boosted trees, is best matched to the resulting 187-dimensional space, and why?

Our contributions are as follows. First, we release a complete feature-extraction pipeline that depends only on packages available through standard public indices (opencv-python, scikit-image, scikit-learn, numpy), replacing the closed-source fractal-descriptor toolkit used in [1] with three descriptors whose definitions are fully public. Second, we report the first classification results, rather than a distributional test alone, for the PCA-residual representation on an AI-generated-face task, evaluating five classifiers under an identical, leakage-controlled protocol and identifying which classifier family is best suited to this feature space and why. Third, we situate the pipeline within a synthesis of fifteen related sources spanning fractal, frequency, texture, and noise-based detection, showing precisely where the open handcrafted-feature approach and the PCA-residual representation have, and have not, previously been combined. Fourth, we state explicitly which extensions remain open, a sensitivity sweep over the residual truncation level, a controlled ablation isolating the effect of the PCA-residual step itself, and robustness testing under common image perturbations, so that the boundary between what this paper establishes and what future work must still show is never left ambiguous.

2 Related Work

Deep detectors trained end to end on pixels are the dominant approach to AI-generated face detection, and FaceForensics++ [3] established both the standard benchmark protocol and a widely used Xception baseline for this setting. Such networks reach strong in-distribution accuracy, but their reliance on millions of learned parameters makes them costly to train and, because the decision boundary is distributed across the network rather than attached to a small set of named quantities, difficult to explain to a non-specialist audience or defend feature-by-feature. This has kept a second line of work active alongside deep detectors: fixed, hand-

designed descriptors paired with a classical classifier, which trades some raw accuracy for a model that is cheap to train, fast at inference, and fully inspectable.

A specific finding motivates the frequency-domain half of our feature vector. Zhang, Karaman, and Chang show that the upsampling and convolution operations inside common GAN generators leave a detectable, structured artifact in the frequency spectrum of a generated image, and that a simple frequency-domain classifier trained on simulated artifacts already generalizes across several GAN architectures [7]. Tan et al. build directly on the premise that frequency information is where synthesis artifacts concentrate, proposing a frequency-domain network, FreqNet, and explicitly positioning frequency- and spectrum-based representations as a lightweight, interpretable alternative to raw-RGB convolutional pipelines that remains competitive with heavier baselines [14]. Both results support treating a frequency-magnitude histogram as an informative, non-arbitrary descriptor rather than an ad hoc choice.

A parallel body of work supports the texture and noise half of our feature vector, and also supplies the closest methodological precedents for the overall recipe of concatenating several handcrafted descriptors and feeding them to a classical classifier. Local binary patterns, in the rotation-invariant uniform formulation we use, were introduced by Ojala, Pietikäinen, and Mäenpää as a general-purpose, mathematically complete texture descriptor for gray-scale images [15]. The noise-residual half is supported by Cozzolino and Verdoliva’s Noiseprint, which shows that camera and synthesis pipelines leave a residual noise fingerprint distinguishable after suppressing scene content [13]. Three recent studies apply this handcrafted-plus-classical-ML recipe directly to face forgery detection: Yasir and Kim fuse HOG, LBP, and KAZE descriptors on raw face images and train an ensemble of RandomForest, XGBoost, ExtraTrees, and SVC classifiers, reporting 92 percent accuracy on FaceForensics++ and 96 percent on Celeb-DFv2 [4]; Nirob et al. fuse a comparable set of handcrafted descriptors with a LightGBM classifier on the CIFAKE benchmark [9], and Chaudhari et al. combine HOG, LBP, and facial landmark descriptors with logistic regression for the same task [10]. These three studies confirm that handcrafted-feature-plus-classical-classifier pipelines are an active, competitive approach in 2025 and 2026, not a historical curiosity. Our pipeline differs from all three in operating on the PCA residual of the image rather than on the raw image directly.

That residual representation has its own, older lineage that predates the specific fractal formulation in [1] by roughly a decade. Yang, Peng, Li, and Long extract noise-standard-deviation and lacunarity-texture statistics for image tamper detection with an SVM classifier as early as 2016 [11], establishing that the combination of a noise estimate with a fractal-texture statistic (lacunarity) under a classical classifier is a long-standing, previously validated recipe, just not one previously applied to the PCA-residual-of-a-face-image setting that [1] introduces. Giudice, Guarnera, and Battiato analyze DCT-domain statistics for GAN-image detection and, notably, explicitly test their detector’s robustness to JPEG compression and resizing [12], which is the kind of robustness protocol we identify as necessary future work for our own pipeline in Section 6.3. Within the specific fractal-descriptor family used by [1], Xiao et al. detect AI-generated images via a different mechanism, fractal self-similarity measured directly in the frequency spectrum rather than in a PCA residual [2]; Wang et al. also build on fractal structure but for proactive watermark-based localization rather than passive detection, addressing a different problem [5]; and Mohan and Peebles show that lacunarity, one of the three fractal descriptors used in [1], can be reformulated as a differentiable, learnable pooling layer inside a neural network rather than computed once and handed to a fixed classifier [6], which is the concrete mechanism by which a future version of this line of work could move from a fixed-recipe contribution to an architectural one. Finally, Hinke-Navarro et al. show that fusing several handcrafted transforms, SRM, DCT, ELA, and SVD among them, with a deep RGB stream remains an active 2025 direction, indicating that handcrafted and deep representations are increasingly treated as complementary rather than competing

[8].

Positioned against this literature, the present paper occupies a specific, previously empty cell: it takes the PCA-residual representation validated by [1] and, instead of pairing it with a closed-source fractal toolkit, pairs it with the open handcrafted-feature-plus-classical-ML recipe validated by [4], [9], and [10] on raw images. No source we identify applies that recipe to the PCA residual, and no source in the PCA-residual line of work reports a classifier accuracy rather than a distributional test. Closing that specific gap, not proposing a new architecture, is the contribution of this paper: a recipe-level extension of an existing representation, not a new model or problem formulation.

3 Problem Formulation and Preliminaries

The task is binary classification: given a grayscale face image, decide whether it is a real photograph ($y = 0$) or an AI-generated face ($y = 1$). Every image is first resized to a fixed 256×256 grayscale array I .

Rather than classifying I directly, the pipeline first computes a residual image R that removes the part of I explained by a small number of dominant directions. Treating I as 256 row-vectors, each of dimension 256, principal component analysis is fit on those 256 rows, independently for every single image, producing an orthonormal set of row-direction vectors $u_1, u_2, \dots, u_{256}$ ordered by decreasing explained variance and a row mean μ . Each row I_i has coefficients $c_{i,k} = (I_i - \mu) \cdot u_k$ along direction u_k . The residual image keeps every row’s coefficients along the bottom 224 directions and discards, that is, sets to zero, its coefficients along the top 32 directions:

$$R_i = \mu + \sum_{k=33}^{256} c_{i,k} u_k = I_i - \sum_{k=1}^{32} c_{i,k} u_k$$

for every row $i = 1, \dots, 256$, where the truncation level 32 is a fixed hyperparameter of the pipeline, not something tuned inside this paper (see Section 6.3). Because this PCA is refit independently for each image rather than fit once on a training corpus and reused, it introduces no parameter that could leak information between the training, validation, and test splits. The residual of a test image never depends on any other image in the dataset.

A feature map $\phi : R \mapsto \mathbb{R}^{187}$ then converts the residual image into a fixed-length vector by concatenating three histogram-based descriptors, defined precisely in Section 4. Given a labeled training set $\{(\phi(R^{(j)}), y^{(j)})\}$, the goal is to fit a classifier $f_\theta : \mathbb{R}^{187} \rightarrow [0, 1]$ that estimates $P(y = 1 \mid \phi(R))$ and thresholds it into a real/fake decision, evaluated by held-out accuracy, F1, and AUC.

4 Method

4.1 Pipeline overview

The pipeline has five stages, applied identically to every image regardless of split: (i) load the image and convert it to grayscale at 256×256 resolution; (ii) compute the PCA residual R as defined in Section 3, independently for that single image; (iii) compute three histogram descriptors on R , not on the original image; (iv) concatenate the three histograms into a single 187-dimensional vector; (v) standardize each of the 187 dimensions to zero mean and unit variance using a scaler fit exclusively on the training split, then apply the same fitted scaler to the validation and test splits. Stage (v) is the only stage in the pipeline where

information about the training split’s statistics is allowed to touch the validation or test data, and it touches them only through two numbers per dimension, a mean and a standard deviation, which is the standard, leakage-safe use of a scaler.

4.2 Handcrafted feature extraction

All three descriptors below operate on the residual image R produced in Section 3, not on the original image I . This ordering, residual first, descriptors second, is the central methodological choice of the paper and the one point in the pipeline directly inherited from [1].

Frequency descriptor. The two-dimensional discrete Fourier transform of R is computed, shifted so that the zero frequency is centered, and its magnitude is compressed with a log transform, $m = \log(1 + |\mathcal{F}(R)|)$, to prevent the small number of very large low-frequency coefficients from dominating the histogram. The 65,536 pixel values of m are then binned into a 64-bin histogram spanning that image’s own minimum and maximum log-magnitude, and the histogram is normalized to sum to one. This descriptor is included because upsampling and convolutional layers inside GAN and diffusion decoders are known to leave structured, non-photographic artifacts in the frequency magnitude spectrum [7], and because frequency-domain summaries of exactly this kind have been shown to remain informative and lightweight relative to raw-pixel networks [14].

Texture descriptor. A local binary pattern is computed on R with $P = 8$ sampling neighbors at radius $\rho = 1$, using the rotation-invariant uniform variant `nri_uniform` of Ojala et al. [15]. This variant assigns every pixel one of $P(P - 1) + 3$ possible codes, which for $P = 8$ is exactly $8 \times 7 + 3 = 59$ codes. The resulting per-pixel code map is binned into a 59-bin histogram covering that entire code range and normalized to sum to one. The choice of `nri_uniform` rather than the plainer uniform variant is deliberate and is defended in full in Section 4.4, because it is exactly the kind of substitution a reviewer or an examiner is likely to probe.

Noise descriptor. A high-pass noise estimate is obtained by Gaussian-blurring R with a 5×5 kernel and subtracting the blurred image from R itself, $n = R - \text{GaussianBlur}_{5 \times 5}(R)$. This is a second-order residual: R already removed the dominant PCA structure of the original image, and n further removes the locally smooth structure of R , isolating pixel-to-pixel variation at the finest scale the pipeline examines. The values of n are binned into a 64-bin histogram over the fixed range $[-50, 50]$, chosen because n is a bounded difference of two 8-bit-range images, and normalized to sum to one. This descriptor is motivated by evidence that camera sensor noise and generative-model synthesis noise carry distinguishable, low-level residual signatures [13].

Concatenation. The three histograms are concatenated in a fixed order, frequency then texture then noise, giving a $64 + 59 + 64 = 187$ -dimensional raw feature vector for every image, which is then standardized as described in Section 4.1.

4.3 Feature-extraction algorithm

Algorithm 1 PCA-Residual Handcrafted Feature Extraction

```
Require: grayscale image I of size 256x256; number of components to
        remove N = 32; LBP parameters P = 8, radius = 1,
        method = nri_uniform
Ensure:  standardized feature vector phi(I) in R^187

1:  fit a per-image PCA on the 256 rows of I, giving row mean mu
    and ordered directions u_1, ..., u_256
```

```

2:  for each row  $I_i$  of  $I$ , compute coefficients  $c_{\{i,k\}} = (I_i - \mu) \cdot u_k$ 
3:  zero out the coefficients for  $k = 1, \dots, N$  (discard the  $N$ 
    dominant directions) and reconstruct each row from the rest
4:   $R \leftarrow$  reconstructed image                # the PCA residual
5:   $m \leftarrow \log(1 + \text{abs}(\text{FFT2D}(R)))$ , shifted to center zero frequency
6:   $h_{\text{freq}} \leftarrow$  normalized 64-bin histogram of  $m$  over  $[\min(m), \max(m)]$ 
7:   $L \leftarrow \text{LocalBinaryPattern}(R, P, \text{radius}, \text{method} = \text{nri\_uniform})$ 
8:   $h_{\text{tex}} \leftarrow$  normalized 59-bin histogram of  $L$  over  $[0, 59)$ 
9:   $n \leftarrow R - \text{GaussianBlur}_{5 \times 5}(R)$ 
10:  $h_{\text{noise}} \leftarrow$  normalized 64-bin histogram of  $n$  over  $[-50, 50]$ 
11:  $v \leftarrow \text{concat}(h_{\text{freq}}, h_{\text{tex}}, h_{\text{noise}})$           # 187-dim raw vector
12:  $\text{phi}(I) \leftarrow (v - \text{mean\_train}) / \text{std\_train}$       # per-dimension
    # standardization,
    # statistics fit on
    # the training split
13:  return  $\text{phi}(I)$ 

```

Algorithm 1 reads as follows. Lines 1 through 4 are the reason this pipeline exists at all: instead of handing the classifier the raw pixels of the image, we first strip away the part of the image that a 32-direction linear model of its own rows already explains, and we do this per image so that no cross-image statistic and no split boundary can leak into the residual. The motivation, inherited directly from [1], is that a real camera photograph and a synthetically generated face can look nearly identical in their dominant, large-scale structure, faces are faces, but the finer structure that survives after that dominant structure is removed is where the two populations are hypothesized, and were shown in [1] under a different descriptor, to separate.

Lines 5 and 6 ask a narrower question of the residual: does its energy sit at the frequencies a generator’s own architecture would be expected to disturb? Generators built from transposed convolutions and upsampling layers imprint a partially periodic structure on the frequency magnitude of their output [7]. Taking the log first, in line 5, is a standard step that prevents the handful of very large low-frequency coefficients present in almost any natural image from swamping the 64-bin histogram and drowning out the higher-frequency bins where the artifact of interest is expected to live.

Lines 7 and 8 ask a different, texture-level question: independent of frequency content, does the local micro-pattern of pixel neighborhoods look like the micro-pattern of a photographed surface? The specific choice of `nri_uniform` in line 7, rather than the more commonly seen uniform LBP variant, is not cosmetic. For $P = 8$ neighbors, the uniform variant is conventionally binned into $P + 2 = 10$ categories, nine “uniform” rotation patterns plus one catch-all “non-uniform” bin. Forcing that 10-category output into a 59-bin histogram, as an earlier, uncorrected version of this pipeline did, leaves the great majority of the 59 bins, up to 49 of them, structurally unreachable no matter what the image contains, silently discarding on the order of a quarter of the total 187-dimensional feature space before the classifier ever sees it. The `nri_uniform` variant instead defines exactly $P(P - 1) + 3 = 59$ genuinely distinct, reachable codes for $P = 8$, so every one of the 59 bins in line 8 corresponds to a code the descriptor can actually produce. This is the single most defensible engineering decision in the pipeline, precisely because it has a wrong alternative that looks superficially reasonable, produces a histogram of the same declared size, and yet quietly encodes far less information. We verified directly, by inspection of the extraction code, that the pipeline whose results are reported in Section 5 uses `nri_uniform`, not `uniform`.

Lines 9 and 10 ask a third, independent question: at the very finest pixel-to-pixel scale, after both the dominant PCA structure and the local texture pattern have already been accounted for by the previous two de-

scriptors, is there a residual noise character that differs between a captured photograph’s sensor noise and a generator’s synthesis noise [13]? Subtracting a mildly blurred copy of R from itself is a standard, parameter-light way to isolate that finest scale without introducing another learned or tuned component.

Line 11 concatenates the three histograms in a fixed, always-identical order, and line 12 standardizes every one of the 187 dimensions using only statistics computed on the training split. This last step matters because the three descriptors live on different natural scales, a normalized frequency histogram, a normalized texture histogram, and a normalized noise histogram, all before standardization sum to one internally but are not on a comparable per-bin scale with one another, and several of the five classifiers evaluated in Section 5, most directly the linear one, are sensitive to exactly that kind of scale mismatch.

4.4 Classifiers

Five classifiers are trained on the identical standardized 187-dimensional vector, with no per-classifier feature engineering: a thresholded linear regression, used as the simplest possible linear baseline; a RandomForest, an unboosted bagged-tree ensemble; and three boosted-tree implementations, XGBoost, LightGBM, and CatBoost, which differ from one another mainly in how they grow trees and regularize splits rather than in the objective they optimize. All five were trained with each library’s standard default hyperparameters. No venue-specific hyperparameter search was run for any of the five, which we treat as an explicit scope limitation rather than an oversight, since a tuned comparison could change the ranking reported in Section 5 and should not be read as ruled out by this paper.

5 Experimental Setup

Dataset. The dataset contains 40,000 face images, balanced between 20,000 real images sampled from FFHQ, Flickr-Faces-HQ, a dataset of photographic faces introduced by Karras, Laine, and Aila [16], and 20,000 AI-generated images sampled from the StyleGAN-generated “1 Million Fake Faces” collection distributed on Kaggle [17]. Every image is converted to grayscale and resized to 256×256 pixels before any further processing.

Split. Images are split 80/10/10 into training, validation, and test sets using stratified sampling with a fixed random seed (42), first separating 80 percent for training and 20 percent for a temporary pool, then splitting that pool evenly into validation and test. This yields 32,000 training images, 4,000 validation images, and 4,000 test images, each split balanced between the two classes by construction. The StandardScaler used in Section 4.1 is fit exclusively on the 32,000 training vectors and then applied unchanged to the validation and test vectors, so no statistic derived from validation or test data ever influences a value the classifiers are trained on.

Truncation level. The PCA truncation level is fixed at $N = 32$ removed components for every image, matching the default value used in [1]. This value is inherited from the base study, not independently tuned or swept inside this paper. Section 6.3 states this as an explicit limitation rather than presenting $N = 32$ as an optimized choice.

Model selection. For each of the five classifiers in Section 4.4, the validation split is used only to confirm that a classifier trained on the training split behaves as expected before it is scored once, and only once, on the held-out test split. No classifier’s hyperparameters were adjusted based on a test-set score, and no test-set result influenced the choice of which classifier to report as best.

Reproducibility. The full pipeline, preprocessing, PCA-residual computation, feature extraction, and training of all five classifiers, was executed end to end and independently re-run once on the same 40,000-image dataset and the same fixed seed. The two runs agreed to within ordinary floating-point and library-version variation, and the numbers reported in Section 6 are taken from that verified pipeline. No result in this paper is estimated, interpolated, or carried over from a different dataset or a different split.

6 Results and Discussion

6.1 Overall classification performance

Table 1 reports validation and test accuracy, F1, and AUC for all five classifiers. Every classifier beats a trivial 50 percent baseline by a wide margin, confirming the central empirical claim of the paper: the PCA-residual representation, when paired with open frequency, texture, and noise descriptors rather than the closed-source fractal descriptors used in [1], still carries enough real/fake signal to support a genuine classifier, not merely a distributional test.

Table 1. Validation and test performance of all five classifiers on the 187-dimensional PCA-residual feature vector.

Classifier	Val Acc	Val F1	Val AUC	Test Acc	Test F1	Test AUC
Linear regression (thresholded)	0.8535	0.8556	0.9321	0.8552	0.8574	0.9309
RandomForest	0.8355	0.8346	0.9121	0.8350	0.8338	0.9144
XGBoost	0.8680	0.8685	0.9415	0.8688	0.8687	0.9442
LightGBM	0.8690	0.8700	0.9424	0.8662	0.8664	0.9427
CatBoost	0.8602	0.8608	0.9360	0.8598	0.8594	0.9386

Validation and test numbers are close to one another for every classifier, typically within half a percentage point of accuracy and within roughly 0.003 of AUC, which is consistent with, though not formal proof of, the split not having leaked information and the feature space being reasonably stable across random partitions of the same 40,000-image pool.

6.2 Which classifier family wins, and why

The two boosted-tree implementations, XGBoost and LightGBM, are essentially tied for best and clearly ahead of RandomForest and CatBoost: XGBoost reports the best test accuracy at 0.8688, the best test F1 at 0.8687, and the best test AUC at 0.9442, with LightGBM a close second on all three test metrics despite reporting the best validation accuracy, F1, and AUC of the two. We report both rather than declaring a single winner because the gap between them, on the order of 0.002 to 0.003 across every test metric, is well within the range that a different random seed or a different library minor version could plausibly move, and nothing in our protocol licenses treating a difference that small, or a ranking that flips between validation and test, as a demonstrated ranking.

A more interesting and more stable pattern is the gap between the boosted-tree pair and RandomForest, which trails by roughly three accuracy points and a full 0.03 of AUC despite also being a tree ensemble. Unlike XGBoost, LightGBM, and CatBoost, RandomForest’s individual trees are grown independently and averaged rather than grown sequentially to correct one another’s residual errors. This suggests that the 187-dimensional feature space, while informative, contains its class-separating signal in interactions among the

frequency, texture, and noise histograms that are easier for sequential, error-correcting boosting to isolate than for independently averaged bagged trees.

The thresholded linear regression baseline is the most informative single number in Table 1, precisely because it is the simplest model and still reaches 0.8552 test accuracy and 0.9309 test AUC, within roughly one and a half accuracy points of the best boosted tree. A purely linear decision boundary over the 187 standardized histogram bins already separates the two classes well, which indicates that a large share of the real/fake signal in this feature space is close to linearly separable, and that the gains from tree-based non-linearity, while real and consistent, are a refinement of an already strong linear signal rather than the source of most of the separability. We read this as evidence that the descriptors themselves, not classifier sophistication, are doing most of the work, which is the outcome an interpretable, feature-driven pipeline should aim for.

6.3 Caveats and threats to validity

Several caveats bound what this paper’s numbers support, and we state them here rather than leaving them implicit. The PCA truncation level $N = 32$ was inherited from [1] and never swept inside this paper. We do not know whether a smaller or larger truncation would move accuracy up or down, and no claim in this paper should be read as depending on 32 being an optimal choice. We did not run a controlled ablation comparing the same three descriptors extracted from the PCA residual against the same three descriptors extracted from the raw, non-residual image, so while the pipeline is motivated by the residual step, this paper does not itself provide the isolated, single-variable comparison that would demonstrate the residual step’s individual contribution. That comparison is scoped as immediate future work in Section 7. No robustness test under common perturbations, JPEG recompression, blur, additive noise, or resizing, was performed, so the reported numbers describe performance on clean images from the same acquisition and generation pipelines used at training time, and should not be read as a claim about performance under adversarial or degraded conditions. We did not fine-tune a deep convolutional baseline, such as Xception, on this exact 40,000-image split. The 92 to 96 percent figures reported by [4] and the strong results reported by [3] and [9] are cited in Section 2 as context for where handcrafted-feature approaches sit relative to the wider literature, not as a controlled, same-split comparison, since those figures come from different datasets and different splits than the one used here. The reproducibility claim in this section and in the paper’s title covers the data split and the per-image PCA computation, both fixed by the seed described in Section 5. It does not extend to the five classifiers’ own internal randomness, for example RandomForest’s bootstrap resampling, which was left at each library’s default and was not separately seeded or version-pinned. An exact digit-for-digit reproduction of Table 1 on a different machine or library version is therefore not guaranteed, only reproduction within the ordinary variation already described in Section 5. Finally, the accuracy, F1, and AUC values in Table 1 are point estimates from a single held-out test split rather than an average over repeated splits or bootstrap resamples, so we report them without a confidence interval and note explicitly that we do not have the statistical evidence to state how much the ranking in Section 6.2 would vary under resampling.

7 Conclusion

This paper shows that the PCA-residual representation introduced for AI-generated face detection in [1], previously validated only through a distributional test over closed-source fractal descriptors, still separates real from AI-generated faces when paired with three open, publicly documented descriptors, a frequency-magnitude histogram, a rotation-invariant local binary pattern histogram, and a Gaussian high-pass noise histogram, and a classical classifier. Trained on 40,000 balanced grayscale faces and evaluated on a 4,000-

image held-out test set, gradient-boosted trees reach 86 to 87 percent accuracy and an AUC above 0.94, with a purely linear classifier already reaching within roughly one and a half accuracy points of that result, indicating that most of the separating signal lives in the descriptors themselves rather than in classifier complexity. We verified, against the pipeline’s own extraction code, that the texture descriptor uses the mathematically complete `nri_uniform` local binary pattern variant rather than the uniform variant that would silently leave most of its 59 bins unreachable, and we walked through the reasoning behind every stage of the feature-extraction algorithm so that each design choice can be defended individually rather than treated as a black box. Future work includes a sensitivity sweep over the PCA truncation level and a correlation analysis against the Kolmogorov-Smirnov statistic reported by [1], a controlled ablation isolating the effect of the PCA-residual step by comparing identical descriptors extracted from residual versus raw images, robustness testing under JPEG recompression, blur, additive noise, and resizing following the protocol demonstrated by [12], and, at a longer horizon, replacing the fixed lacunarity-style descriptor with a learnable pooling formulation as demonstrated by [6], which is the concrete mechanism by which this line of work could move from a fixed-recipe contribution to an architectural one. All code, the feature-extraction pipeline, and the trained-model evaluation script are available from the authors. Every number reported in this paper comes from that code, executed on the data described in Section 5.

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